

Tilitoimistojärjestelmä

Finnish SaaS Accounting Office System — Full System Design

Version	Date	Deployment
1.0 — Initial Design	May 2026	SaaS Multi-tenant

Methodology	Codebase Est.	Timeline
Test-Driven Development	~211,000 LOC	18–24 months

Tilitoimistojärjestelmä — Architecture & Design Plan

Context

Building a SaaS-based Finnish accounting office system from scratch (greenfield repo). The platform serves accounting firms (tilitoimistot) as primary tenants — each firm subscribes and manages their own client companies (asiakkaat) as isolated sub-tenants. Accountants work on web; customers get a lightweight mobile+web portal. Must comply fully with Finnish accounting law, tax authority integrations, and GDPR.

User answers: - Deployment: SaaS (multiple tilitoimistot, platform-as-a-service) - Payroll: In-house Finnish engine (not 3rd party) - Backend stack: Architecture-driven recommendation (see below) - Mobile scope: Customer portal only (scan receipts, approve invoices, view reports) - Development approach: **Test-driven development (TDD)**

Tenancy Model

```
Platform (SaaS operator)
├─ Tilitoimisto / Tenant  [1..N]
│   └─ Asiakas / Sub-tenant  [1..M per tilitoimisto]
```

Isolation: PostgreSQL Row Level Security (RLS) on every table. -
tenant_id + asiakas_id columns everywhere - RLS policies:
platform_admin sees all; tiltoimisto users see their asiakkaat;
asiakas users see only their own data - RLS must be tested with a
dedicated security test suite before any production deployment

Technology Stack

Layer	Technology	Justification
Backend API	Python 3.12 + FastAPI	Decimal precision for finance, strong typing (Pydantic), async I/O, best OCR/ML ecosystem
Database	PostgreSQL 16 + RLS	ACID, RLS for tenant isolation, JSONB for document metadata, pg_cron for scheduled jobs
Task queue	Celery + Redis	Async OCR, OmaVero submissions, bank sync, tulorekisteri
Web frontend	React 18 + TypeScript + Vite	Large ecosystem, shared component library between tilitoimisto and asiakas apps
Mobile	React Native + Expo	Customer portal only; camera, offline queue, push notifications
Object storage	S3-compatible (MinIO local, AWS S3 prod)	Tenant-scoped keys, immutable archive policy
Auth	Keycloak (self-hosted)	OIDC/OAuth2, MFA, Suomi.fi tunnistus integration
Infra	Kubernetes + Helm + Terraform	Multi-tenant scale, isolated namespaces per environment
CI/CD	GitHub Actions	Test gates before every merge
Monitoring	Prometheus + Grafana + Sentry	Financial system uptime and error tracking

Codebase Size Estimate (TDD ratio)

Layer	Production LOC	Test LOC	Total
Backend (Python)	~60,000	~55,000	~115,000
Web frontend (TS/React)	~40,000	~25,000	~65,000
Mobile (React Native)	~12,000	~8,000	~20,000
Infra / config / SQL	~8,000	~3,000	~11,000
Total	~120,000	~91,000	~211,000

TDD raises test LOC to ~75% of production code. Financial calculation modules (payroll, ALV) aim for near 100% coverage. This is a medium-large enterprise codebase — comparable to ~2-3 years of focused development by a 4-6 person team.

TDD Strategy

TDD is the right approach here because: - Payroll deduction errors = legal and financial liability - ALV rate errors = OmaVero penalties - Double-entry imbalance = corrupted books - Tenant isolation bugs = catastrophic data leak

Testing layers:

Unit tests	– pure financial calculations (payroll, ALV, double-entry bal
Integration tests	– API endpoints, DB queries, RLS enforcement
Contract tests	– OmaVero API, Tulorekisteri API, Finvoice XML schema
E2E tests	– full user flows (tilitoimisto onboards asiakas, posts tosit
Security tests	– dedicated RLS penetration suite (cross-tenant access attempt

TDD rules for this project: - All financial calculation functions: write test with known Finnish tax scenario BEFORE implementation - ALV rates and payroll rates live in test fixtures, not hardcoded — tests break if rates not updated - Every OmaVero/Tulorekisteri integration has a mock + a contract test against sandbox API - RLS tests: automated suite attempts cross-tenant data access after every migration

Core Data Model

Tenancy tables

```
organizations      -- tilitoimistot (tenants)
  id, name, y_tunnus, subscription_plan, created_at

org_users          -- tilitoimisto staff
  id, org_id, user_id, role

customers          -- asiakkaat (sub-tenants)
  id, org_id, name, y_tunnus, tilikausi_alkaa, tilikausi_loppuu, alv_jakso

customer_users     -- asiakas portal users
  id, customer_id, user_id, role
```

Kirjanpito core

```
tilikartat         -- chart of accounts per asiakas
tilit              -- accounts: numero, nimi, tyyppi, alv_koodi
tilikaudet         -- fiscal years: alkaa, loppuu, tila (avoin/suljettu/lukittu)
tositteet          -- vouchers: tyyppi, pvm, numero (auto), tila
viennit            -- journal lines: tili_id, debet, kredit, alv_% (debet=kredit)
```

Reskontra

myyntilaskut	-- AR invoices + Finvoice metadata
ostolaskut	-- AP invoices + approval state machine
suoritukset	-- payments matched to invoices
tilitapahtumat	-- bank statement lines (from CAMT.053/MT940)

Palkanlaskenta

tyontekijat	-- employees: henkilotunnus (encrypted), ennakonpidatys%,
palkkakaudet	-- pay periods
palkat	-- calculated pay: brutto, ennakonpidatys, tyel, sairausva
palkka_verokortti	-- withholding taxcard per employee per year
tulorekisteri_log	-- submission log + API responses
alv_vuosikorot	-- ALV rate table by year (NOT hardcoded)
palkkavakuutusmaksut	-- insurance rate table by year (NOT hardcoded)

ALV & Authorities

alv_kaudet	-- VAT periods per asiakas
alv_laskelmat	-- VAT calculation lines by code
omavero_lahetykset	-- OmaVero submission log

Archive & Compliance

asiakirjat	-- immutable document registry: s3_key, sha256, retention_
audit_log	-- append-only: user, action, before/after JSON, timestamp

Module Breakdown

Module 1 — Auth & IAM

Roles: platform_admin, tilitoimisto_admin, kirjanpitaja, asiakas_admin, asiakas_user, tilintarkastaja - JWT + refresh tokens via Keycloak - RBAC enforced at API gateway + service level - MFA mandatory for all accountant roles - Suomi.fi tunnistus (Phase 2 option) - All auth events written to audit_log

Module 2 — Kirjanpito (Bookkeeping Core)

- Tilikartta from KILA standard template (customizable)
- Tilikausi state machine: avoin → suljettu → lukittu
- Double-entry enforced: `sum(debet) = sum(kredit)` per tosite (DB constraint)
- Tosite auto-numbering per kausi (YYYY + sequential)
- Period locking: no backdated posts to locked kausi
- Finnish ALV codes (1-99 series) on viennit

Module 3 — Myyntireskontra (AR)

- Myyntilasku with line items and ALV breakdown
- Finnish viitenumero generation (mod 97)
- Finvoice 3.0 XML + PEPPOL BIS 3.0 generation
- Auto-match incoming payments by viitenumero
- Asiakasrekisteri with YTJ lookup by Y-tunnus

Module 4 — Ostoreskontra (AP)

- Receive e-invoices via Finvoice operator (Maventa/Apix broker)
- OCR parse paper invoices (Tesseract)
- Approval state machine: `saapunut → odottaa_hyvaksyntaa → hyväksytty → maksettu → kirjattu`
- SEPA PAIN.001 payment file generation
- Auto-post to kirjanpito after approval

Module 5 — ALV

- Rate table in DB per year (Sept 2024: standard changed 24% → 25.5%)
- Finnish ALV codes: 0%, 10%, 14%, 25.5% (standard), EU reverse charge, OSS
- Period calculator (kuukausittain / neljännesvuosittain based on liikevaihto)
- OmaVero API push with certificate-based OAuth2
- Correction submissions for late/amended periods

Module 6 — Palkanlaskenta (HIGHEST COMPLEXITY)

Finnish deductions (2024, all stored in rate tables): -

Ennakonpidätys — from individual verokortti (fetched from OmaVero or entered manually) - TyEL työntekijä — 7.15% (<53yo), 8.65% (53-62yo) - Sairausvakuutusmaksu — 1.16% (employee) - Työttömyysvakuutusmaksu — 0.79% (employee) - Employer: TyEL, sairausvakuutus, tapaturma, ryhmähenkivakuutus

Process flow: 1. Define palkkauskauti → enter työtunnit/ kuukausipalkat 2. Auto-calculate all deductions from rate tables 3. Generate palkkalaskelma PDF 4. Auto-post palkkakirjanpito viennit 5. Submit to Tulorekisteri API (within 5 days of payment day) 6. Generate TyEL ilmoitus to pension company 7. SEPA PAIN.001 for salary disbursement

Rate tables are DB records, not code constants — annual update is an admin workflow.

Module 7 — Tilinpäätös (Year-End Closing)

- Guided checklist workflow with blocking steps
- Tuloslaskelma: kululajipohjainen (default) + toimintokohtainen
- Tase: Finnish standard format
- Liitetiedot templates: mikroyritys / pk-yritys / suuri yritys
- XBRL/iXBRL export (mandatory for large companies from 2024)
- Auto-closing entries (depreciation, accruals)

- Tilikausi locked after tilinpäätös approval — irreversible

Module 8 — Viranomaisintegraatiot

System	Protocol	Purpose
OmaVero	REST + OAuth2 + org cert	ALV, ennakkovero, yhteisöilmoitus
Tulorekisteri	REST + cert	Real-time palkka ilmoitukset
YTJ	REST	Y-tunnus lookup, company data
Finvoice/ PEPPOL	SFTP/AS4 via operator	E-invoice send/receive

All external API calls: retry logic, timeout, circuit breaker, full response logging to `omavero_lahetykset` / `tulorekisteri_log`.

Module 9 — Pankkiintegraatiot

- **Inbound:** CAMT.053 XML + MT940 import → tilitapahtumat table
- **Auto-reconciliation:** match by viitenumero, then by amount+date+toimittaja
- **Outbound:** SEPA PAIN.001 generated from approved ostolaskut + palkat
- Supported banks: Nordea, OP, Danske, Handelsbanken, S-Pankki (all use CAMT.053 now)

Module 10 — Asiakasportaali

Web + Mobile (React Native/Expo): - Dashboard: velat, saatavat, ALV-status, next deadline - Receipt upload: drag-drop (web) / camera scan (mobile) - Invoice approval queue with comment field - Report viewer: P&L, tase, ALV summary (read-only) - Threaded messaging per document with kirjanpitäjä - Offline

receipt queue on mobile (sync on reconnect) - Push notifications (Firebase): "Lasku X odottaa hyväksyntääsi"

Module 11 — Tilitoimiston Master View

- All clients dashboard with RAG status (red/amber/green)
- Deadline tracker: ALV, palkat, tilinpäätös — all clients unified calendar
- Missing receipts count per asiakas
- Kirjanpitäjä workload assignment and view
- Alert engine: "3 clients have ALV due in 5 days"

Module 12 — Raportointi & Arkisto

- Standard reports: tuloslaskelma, tase, pääkirja, päiväkirja, ALV-erittely, reskontra, palkkayhteenveto
- Export: PDF (WeasyPrint), Excel (openpyxl), CSV
- 10-year immutable archive (S3 object lock, WORM policy)
- SHA256 hash stored per document at upload — integrity verifiable
- GDPR: henkilötunnus encrypted at rest with customer-specific key; right-to-erasure handled by key deletion

Module 13 — OCR & Auto-posting

- Pipeline: upload → S3 → Celery worker → Tesseract OCR → extract (toimittaja, pvm, summa, ALV, viite)
- ML tili suggestion: trained on asiakas's own posting history
- Kirjanpitäjä reviews suggestion, accepts or corrects
- Active learning: corrections fed back to model

Module 14 — Notification & Deadline Engine

- pg_cron + Celery beat for scheduled checks
- Deadlines tracked: ALV (12th of following month), Tulorekisteri (5 days post-payment), Tilinpäätös (4 months post-tilikausi)
- Channels: email (SendGrid), push (Firebase), in-app

API Design

- REST (not GraphQL) — financial resources have clear semantics
 - OpenAPI 3.1 spec — contract-first, generate clients
 - Versioned: `/api/v1/...`
 - JWT bearer tokens with `tenant_id`, `asiakas_id`, `role` claims
 - Separate API contexts: `/tilitoimisto/...` (accountant) vs `/portal/...` (customer)
-

Phased Build Roadmap

Phase	Duration	Deliverables
1 — Foundation	3-4 mo	Infra, auth/IAM, tilikartta, kirjanpito core, tilikausi, audit log
2 — Reskontra	2-3 mo	AR, AP, approval workflow, bank import, reconciliation, PAIN.001
3 — ALV & Authorities	2 mo	ALV engine, OmaVero API, YTJ, Finvoice operator
4 — Payroll	3-4 mo	Full palkanlaskenta, Tulorekisteri API, TyEL, palkkakirjanpito
5 — Year-End	2 mo	Tilinpäätös workflow, XBRL, statutory reports, PDF/Excel
6 — Portal & Mobile	2-3 mo	Asiakasportaali web, React Native app, OCR pipeline, master dashboard
7 — Intelligence	ongoing	ML auto-posting, advanced analytics, Suomi.fi tunnistus

Phases 1-3 = usable product. Full system ~18-24 months.

Key Design Decisions (must resolve before coding)

1. **Finvoice operator:** Use Maventa or Apix as broker — do not implement operator directly
 2. **OmaVero cert:** Software cert per tenant stored in secrets manager (HSM for production)
 3. **ALV + payroll rates:** Never hardcoded — DB rate tables with annual admin update workflow. Tests will fail if rates are stale.
 4. **Tilikausi lock:** Enforced at DB level (trigger), not just application layer
 5. **Henkilötunnus:** Encrypted at rest, never in logs, access restricted to payroll module only
 6. **Mobile:** Customer portal only — kirjanpitäjät on web exclusively
 7. **Tilikartta:** KILA standard as seed template, customer can extend account numbering but cannot break standard group ranges
-

Complexity & Risk Assessment

Module	Complexity	Risk	Key Mitigation
Auth & RLS	Medium	Critical	Dedicated security test suite; pentest before launch
Kirjanpito core	Medium	Medium	DB constraint for double-entry; TDD on all posting logic
Myyntireskontra	Medium	Low	Standard; Finvoice schema has XSD validation
Ostoreskontra	Medium	Low	Approval state machine tested exhaustively
ALV engine	High	High	TDD with known Verohallinto scenarios; rate table tests
Palkanlaskenta	Very High	High	TDD first; dedicated team; rate tables tested annually
Tulorekisteri	High	High	Real-time mandate; test against sandbox API; retry + idempotency
OmaVero integration	High	High	Certificate management; sandbox testing; response logging
Bank integrations	High	Medium	Format coverage tests with real bank sample files
Tilinpäätös / XBRL	High	Medium	

Module	Complexity	Risk	Key Mitigation
			XBRL taxonomy well-documented; validate against Eurofiling
OCR / ML	High	Low	ML accuracy improves over time; human review always available
Multi-tenancy RLS	High	Critical	Automated cross-tenant penetration tests on every migration
GDPR vs retention	Medium	High	Legal review needed — right-to-erasure vs. 10-year accounting law conflict

Verification Plan

- **Unit:** All financial functions (ALV calc, payroll deductions, double-entry balance) — pytest with parameterized Finnish tax scenarios from Verohallinto published examples
- **Integration:** API endpoint tests (httpx + pytest) with real PostgreSQL including RLS
- **Contract:** OmaVero sandbox API + Tulorekisteri test environment — validate XML schemas
- **Security:** Automated suite: login as asiakas_user of Tenant A, attempt to read Tenant B data — must return 403/empty at every endpoint
- **E2E:** Playwright (web) + Detox (mobile) — core flows: onboard asiakas, post tosite, submit ALV ilmoitus, run payroll, view tilinpäätös

- **Compliance:** Annual review against Kirjanpitolaki updates and Verohallinto guideline changes

Public API Integration Reference

All APIs verified against current public documentation. Sandbox environments confirmed available for all critical integrations.

1. Verohallinto / Vero API

Purpose: ALV ilmoitus, ennakkovero, verokortti fetch, ennakonpidätys rates

Property	Detail
Auth	mTLS client certificate (PKCS#12) + <code>vero-softwarekey</code> header (software-specific key)
Base URL	<code>api-developer.vero.fi</code> (sandbox), production URL via vero.fi registration
Format	JSON (primary), XML for some interfaces
Encoding	UTF-8
Sandbox hours	08:00–16:00 weekdays only (maintenance outside hours)
Cost	Free
SDK	None official — direct REST calls; contact <code>ohjelmistotalot@vero.fi</code> for dev support

Key operations: - `POST` ALV-ilmoitus (VAT return submission) - `GET` verokortti (employee withholding tax card) — fetched per henkilötunnus - `POST` ennakkovero payments - `GET`

ennakonpidätystiedot (withholding tax info for employer) - POST
yhteisöilmoitus (annual corporate tax return)

Implementation notes: - One certificate per tilitoimisto tenant (stored in secrets manager) - Certificate issued by Verohallinto after registration — ~2-4 week lead time - All requests logged to omavero_lahetykset with full response body - Circuit breaker: if sandbox is outside 08:00–16:00, mock responses in tests

2. Tulorekisteri (Incomes Register) API

Purpose: Real-time palkka ilmoitukset (mandatory within 5 days of payment)

Property	Detail
Auth	X.509 certificate-based (separate certs for: earnings submission, benefits submission, data retrieval)
Submission methods	Real-time web service (per-report), Deferred web service (batch), SFTP
Format	XML only — no BOM character allowed, UTF-8
Sandbox	Stakeholder testing environment available
SDK	None official — direct calls; major payroll software (Palkkaus.fi) as reference

Key operations: - Submit palkka ilmoitus (wage report) per employee per payment - Submit corrections to previous reports - Cancel submitted reports - Retrieve submitted data (data distribution interface)

Implementation notes: - 3 separate certificates: earnings, benefits, retrieval — all must be provisioned at onboarding - Idempotency critical: duplicate submissions must be detected and rejected - 5-day deadline is calendar days from payment date —

deadline engine must track - Test against stakeholder environment
before any production payroll run - XML schema validation must
happen client-side before submission

3. YTJ / PRH Open Data API

Purpose: Company lookup by Y-tunnus, auto-populate asiakas
details, validate business existence

Property	Detail
Auth	None required — fully public API
Base URL	<code>https://avoindata.prh.fi/opendata-ytj-api/v3</code>
Spec	OpenAPI 3.0.1
Format	JSON
Rate limit	300 requests/minute (shared across all users)
Sandbox	No separate sandbox — test directly against live API (data is public register)
Community SDKs	<code>jannewaren/ytj_client</code> (Python), <code>nikked/ Avoindata.fi-Python-Tools</code>

Key operations: - `GET /companies/{businessId}` — fetch by Y-tunnus - `GET /companies?name=...` — search by name - Daily bulk JSON export available for local caching

Implementation notes: - Cache Y-tunnus lookups locally (data updates daily, no need to hit API per request) - Does NOT include sole proprietorships (toiminimet) — manual entry fallback needed - No email/phone numbers — only register data - Use for KYC: validate that asiakas y_tunnus is a real active business

4. Finvoice 3.0 (Format Specification)

Purpose: Finnish national e-invoicing XML format — mandatory for bank network; standard for B2B

Property	Detail
Type	XML format specification (not an API — delivered via operators)
Maintained by	Finance Finland (Finanssiala ry)
Schema	XSD + Schematron validation rules — publicly downloadable
Encoding	UTF-8 or ISO 8859-15
Latest version	3.0 (implemented Sept 2025+)
EU compliance	Mapped to EN 16931 standard
Spec URL	finanssiala.fi/en/topics/finvoice-standard/

Implementation notes: - Generate Finvoice 3.0 XML in-house — spec is publicly available - Validate against XSD schema before sending to operator - Amounts: support 2-5 decimal places - Delivery: always via operator (Maventa or Apix) — never direct to bank - TEAPPSXML 3.0 is the equivalent operator-to-operator routing format (fully compatible)

5. PEPPOL BIS 3.0

Purpose: EU-wide e-invoicing — required for public sector (B2G) in Finland

Property	Detail
Type	UBL 2.1 XML format over PEPPOL network
Access via	Certified PEPPOL access point (Maventa is access point in Finland)
Documentation	docs.peppol.eu/poacc/billing/3.0/
Validators	invoice-navigator.eu/validator (free, public)
Sandbox	Via Maventa sandbox

Implementation notes: - Use Maventa as both Finvoice operator AND PEPPOL access point (single integration) - Validate UBL 2.1 XML against Schematron rules before sending - Growing adoption: mandatory for public sector procurement; B2B use increasing

6. Maventa (Primary E-invoice Operator)

Purpose: Send/receive Finvoice + PEPPOL invoices; operator network routing

Property	Detail
Auth	OAuth2 — <code>client_id</code> (Company UUID) + <code>client_secret</code> (API key) + <code>vendor_api_key</code> (partner key)
Token endpoint	POST <code>/oauth2/token</code>
REST API docs	<code>documentation.maventa.com/rest-api/</code>
Swagger	<code>swagger.maventa.com</code>
Sandbox	Full sandbox available — requires registration
Scale	100,000+ companies in Finland

Key AutoXChange (AX) API operations: - Create company on Maventa network - POST send invoice (Finvoice 3.0 / TEAPPSXML / PEPPOL BIS) - GET receive invoices from network - Register webhooks for delivery status - Activate networks (PEPPOL, Scan, bank network)

Implementation notes: - One Maventa account per tilitoimisto tenant OR white-label partner account (`vendor_api_key` model) - Partner/vendor integration: single API key per software product, sub-accounts per customer - Pricing: Netstamps (prepaid credits) or monthly invoice model - Webhook registration for invoice delivery status — store in `finvoice_lahetykset` log table

7. Apix Messaging (Secondary / Fallback Operator)

Purpose: Alternative e-invoice operator; fallback if Maventa unavailable

Property	Detail
Auth	SHA-256 HMAC digest — <code>TransferID</code> + <code>TransferKey</code>
Method	<code>PUT</code> to submit documents
Format	ZIP file containing XML data files + PDF attachments
API docs	<code>wiki.apix.fi/rest-api-for-external-usage</code>
Pricing	Transaction-based (Netstamps), no fixed costs
Sandbox	Available

Implementation notes: - Different auth model (SHA-256 digest, not OAuth2) — wrap in dedicated client class - Supports automatic format conversion for recipients - Plan: Maventa as primary, Apix as fallback / for specific customers

8. Bank Open Banking APIs (PSD2)

Purpose: Account information (tilitapahtumat), payment initiation (PAIN.001 alternative)

Nordea

Property	Detail
Auth	OAuth2 + SCA (Nordea ID app)
Portal	<code>developer.nordeaopenbanking.com</code>
Sandbox	Full sandbox, free access
SDK	C#, Java, Python samples available
Scale	6,500+ sandbox users, 500+ live companies

OP Pankki (largest in Finland)

Property	Detail
Auth	OAuth2 + SCA
Portal	<code>op-developer.fi</code>
Sandbox	Available, open to all developers
Integrators	GoCardless, Nordigen, Yapily, Klarna also aggregate OP data

Danske Bank Finland

Property	Detail
Auth	OAuth2 + SCA
Portal	<code>developers.danskebank.com</code>
Sandbox	Restricted in Finland — requires TPP license or Open Banking UK directory membership

Implementation notes: - Prefer CAMT.053 file import for most customers (simpler, no OAuth2 per-customer SCA) - Open Banking API for customers who want real-time bank feed (advanced tier feature) - Consider aggregator (Nordigen/GoCardless) to handle multi-bank SCA complexity - S-Pankki and Handelsbanken: CAMT.053 file upload only (no public PSD2 API)

9. SEPA ISO 20022 Bank Formats

Purpose: Payment file (PAIN.001) and bank statement (CAMT.053) processing

Format	Direction	Purpose
pain. 001.001.03	Outbound	Customer credit transfer initiation (salary + invoice payments)
camt. 053.001.02	Inbound	Bank-to-customer account statement
camt. 052.001.02	Inbound	Intraday transaction report (optional, real-time)

Implementation notes: - CAMT.053: primary bank statement format for all Finnish banks - MT940: legacy format — still supported by some banks, must handle both - PAIN.001: generate per payment batch — group by bank, one file per run - Nordea supports both standard and extended camt.053 — handle both variants - Validation: validate XML against ISO 20022 schema before sending to bank - Finance Finland ISO 20022 Payments Guide 2024 is the implementation reference

API Summary Table

API	Type	Auth	Format	Sandbox	Priority
Verohallinto Vero API	Government	mTLS cert + key	JSON	Yes (08-16 only)	P0
Tulorekisteri	Government	X.509 cert (3x)	XML only	Yes	P0
YTJ / PRH	Government	None	JSON	Live only (free)	P1
Finvoice 3.0	Format spec	Via operator	XML	Via operator	P0
Maventa AX API	Operator	OAuth2	JSON/ XML	Yes	P0
Apix	Operator	SHA-256 HMAC	ZIP+XML	Yes	P1 (fallback)
PEPPOL BIS 3.0	Format spec	Via access point	UBL XML	Via Maventa	P1
Nordea PSD2	Bank	OAuth2 + SCA	JSON	Yes	P2
OP Bank PSD2	Bank	OAuth2 + SCA	JSON	Yes	P2
Danske Bank PSD2	Bank	OAuth2 + SCA	JSON	Restricted (FI)	P2
CAMT.053 / PAIN.001	Bank format	Bank file transfer	XML	Bank dep.	P0

P0 = required for MVP | **P1** = required Phase 3 | **P2** = advanced tier feature

Integration Architecture Pattern

All external API clients follow the same structure:

```
ExternalAPIClient
```

```
├─ authenticate()      - cert load / OAuth2 token fetch + refresh
├─ submit(payload)     - send with retry (exp. backoff, max 3 attempts)
├─ validate(response) - assert schema compliance
└─ log(request, response) → external_api_log table
```

Separate log table per integration: `omavero_log`,
`tulorekisteri_log`, `finvoice_log`, `bank_log` All logs: immutable,
append-only, retained 10 years (same as kirjanpito aineisto).

Critical Files to Create (Phase 1)

```
backend/  
  app/  
    core/          -- config, database, auth, RLS middleware  
    modules/  
      kirjanpito/   -- tositteet, viennit, tilikartta, tilikausi  
      alv/          -- rate tables, period calc, OmaVero client  
      palkanlaskenta/ -- deduction engine, tulorekisteri client  
      reskontra/    -- AR, AP, reconciliation  
      tilinpaatos/  -- year-end workflow, XBRL  
    tests/  
      unit/  
      integration/  
      security/     -- RLS cross-tenant tests  
  
  frontend/  
    apps/  
      tilitoimisto/ -- master view, kirjanpito UI  
      portal/       -- customer web portal  
  
  mobile/          -- React Native / Expo (customer portal)  
  
  infra/  
    k8s/  
    terraform/  
  
  docs/  
    api/           -- OpenAPI 3.1 spec (contract-first)
```